CV in SNSF format: draft for the RGO form (upload 2) and the portal

Personal data

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Education

Period	Degree
2019-2023	PhD in Biomedical Sciences, Global Health track, Faculty of Medicine, University of Geneva. Defended 18 December 2023. Dissertation: “Enhancing HIV Prediction through Machine Learning and Streamlining Research with LiteRev”
2023	I-Teams Medical Innovation Program, Faculty of Medicine, University of Geneva
2018	Financial Risk Manager (FRM), Global Association of Risk Professionals
2004-2006	Master of Advanced Studies (DEA) in Econometrics, University of Geneva. Dissertation: “Additive Models and Local Kernel M-Estimator: Bandwidth Selection and Application to the Implied Volatility Surface” (supervised by Prof. Eva Cantoni)
2000-2004	Master of Science in Quantitative Economics, University of Geneva

Positions held

Period	Position
2019-present	Senior Research Associate, Infectious Diseases and Mathematical Modelling, Institute of Global Health, University of Geneva
2024-present	Lecturer, Statistics and Epidemiology, MAS in Public Health, Faculty of Medicine, University of Geneva

Period	Position
2014-2018	Cofounder / Data Scientist, Market Intuitive Investment Managers, Geneva
2007-2014	Financial Risk Manager / Senior Quantitative Analyst, Steenman Asset Management, Geneva
2007	Responsible for Research, Cantonal Service of Statistical Research and Information (SCRIS), Lausanne

Career breaks: none. Academic age about three years (PhD December 2023), after eleven years in quantitative finance.

Supervision, teaching and service

- Supervision of 8 MAS in Public Health theses since 2024 (completed or ongoing), and mentoring of about 6 junior researchers at the Institute (research assistants and interns).
- Lecturer in statistics and epidemiology, MAS in Public Health, since 2024.
- Occasional reviewer for Springer and PLOS journals.

Funding obtained or co-written

- LiteRev development: CHF 70,000 in total (UNIGE 30,000, Fondation Ernst et Lucie Schmidheiny 10,000, Venture Kick 10,000, Mimosa 20,000), obtained with the group.
- Legionellosis study (BASEC 2026-00324, PI Prof. Olivia Keiser): co-wrote the funding, CHF 30,000 from Services Industriels de Genève and CHF 30,000 from the Office cantonal de l’énergie obtained, CHF 30,000 from the Service du médecin cantonal requested.
- GeoAI4EI (Horizon Europe): wrote the part of the proposal on speeding up literature reviews with LiteRev.
- “100 patients complexes” (SNSF, PI Dr Nicolas Perone): co-authored the grant, CHF 30,000 for the analysis, start January 2027.

Languages

French (mother tongue), English (fluent), Turkish (father tongue), German and Spanish (basic).

The four narrative modules and the ten outputs

The module texts are drafted in `cv-narratives/module-1-knowledge.md` (4 outputs), `module-2-individuals.md` (2), `module-3-community.md` (2), `module-4-society.md` (2); the output list with contribution statements is `cv-narratives/research-output-list.md`. They follow in the .docx. Ng'ambi reads accepted, the Edjinedja review reads submitted.

Module 1: Contributions to the generation of new ideas, tools, methodologies or knowledge

I came to biomedical research after eleven years as a quantitative analyst and risk manager, where my work concerned the prediction of rare, costly events: extreme value estimation for non-Gaussian tails, regime and factor models, stress testing and scenario analysis. That inheritance shapes everything below. My contribution to health research has consistently been to bring methods developed for anticipating discontinuities in one domain to problems in another where the same structure exists but the machinery does not.

Prediction where data are incomplete. My doctoral work asked whether individual health status could be predicted from socio-behavioural information alone, in settings where clinical data are sparse. In *PLoS ONE* (2022) I predicted individual HIV status across East and Southern Africa using gradient-boosted models, and - the part I consider the real contribution - characterised where such models transported between countries and where they did not. That question, of when an estimate obtained in one population is usable in another, has stayed with me and is now the centre of my research programme. Related work extended it to treatment-interruption prediction and to latent structure across populations.

A tool, and then a platform. Reviewing literature at the scale these questions demand was the bottleneck, so I built the instrument. **LiteRev** (*Journal of Medical Internet Research*, 2023, first author) combines natural language processing, dimensionality reduction, clustering and nearest-neighbour retrieval to accelerate the identification and structuring of relevant research. It is used beyond my own work: a recent systematic review of artificial intelligence in emergency medical services used LiteRev in its published methods.

Since 2024 I have developed **LiteRev-Evidence**, which extends this from retrieval into structured quantitative extraction: the full review pipeline in working form, from a natural-language or Boolean query searched live across the open literature, through deduplication, screening and PICO extraction, to provenance-tracked, study-quality-scored extraction

pooled into parameter distributions, with per-scenario living reviews and the modelling components that consume the result. Operational scenarios are elaborated with emergency-medicine partners. It is the working prototype of the validated instrument my research programme proposes to build.

Building it produced the question I now want to answer, and I regard that as the most significant intellectual contribution of the work. The platform will readily pool published estimates into prior distributions - and I cannot establish whether it should. Published effect sizes are selectively reported, estimated under different health systems, and automated extraction is weakest precisely on numerical quantities. Whether such evidence improves forecasting when local data are scarce, or misleads confidently at the moment it matters most, is unresolved and consequential.

Clinical and epidemic knowledge. Alongside the methodological work I have contributed substantive findings, including a comparison of clinical severity between Delta and Omicron sub-lineages in a Swiss tertiary centre (*CMI Communications*, 2024, first author), work that required close engagement with hospital clinical data and with the people who generate it.

Module 2: Contributions to the development of other individuals and research teams

I completed my doctorate in December 2023 and have not yet supervised a doctoral researcher to completion; I would rather say that plainly than present this module as more than it is. What I can show is a trajectory that has already crossed into senior authorship, and the concrete ways I enable other people's research.

Senior authorship. I am last author on a machine-learning classification of self-reported cardiovascular disease history across harmonised multi-country WHO STEPS surveys (2014–2019), led by Wingston Ng'ambi and **accepted for publication**. I supervised the statistical analysis and the code. It is the first paper in which my role was to direct another researcher's analysis rather than to conduct my own, and it marks the transition this grant would consolidate.

Teaching. Since 2024 I have lectured in statistics and epidemiology on the **Master of Advanced Studies in Public Health** at the Faculty of Medicine, University of Geneva. The students are largely practising clinicians and public health professionals, many returning to quantitative methods after years away from them. Teaching this group has changed how I work: methods that cannot be explained to the people who will act on their outputs tend, in my experience, to be methods that are never acted upon. That conviction is now built into my research - the project I am proposing evaluates forecasts on whether

they change decisions rather than on statistical accuracy, and that framing came from the classroom before it came from the literature.

Enabling others' research. The tools I build are, in practice, my main contribution to other people's work. LiteRev was designed so that researchers without natural language processing expertise could conduct structured reviews at scale, and it has been used that way. The systematic review of artificial intelligence in emergency medical services for disasters and health emergencies - led by a doctoral researcher at the Université Marie et Louis Pasteur, with colleagues from HUG emergency medicine and the GESICA consortium - used LiteRev to structure its corpus. My contribution was the methodological infrastructure and guidance on its application, working alongside the first author through the screening and clustering stages. Making a method usable by someone else, in their project rather than mine, is a form of contribution I value and intend to keep prioritising.

Supervision. Since 2024 I have supervised eight MAS theses, completed or ongoing, on applied statistics and epidemiology questions brought by the students from their own services, and I have mentored about six junior researchers at the Institute, research assistants and interns, in data management and analysis.

Collaborative practice. Within the GESICA consortium I work across an unusually heterogeneous team - emergency physicians, dispatch clinicians, computer scientists, engineers and an industrial partner, across two countries and two health systems. Much of the useful work in such a setting consists of translation: making a modelling constraint intelligible to a clinician, and a clinical constraint intelligible to an engineer. I have found this to be where projects succeed or quietly fail.

Where this goes. The grant would establish me as principal investigator of a research line I defined, leading its scientific programme, and would give me the standing to supervise doctoral students funded through other channels. The senior authorship above is the first completed step of that transition; the grant is what turns a step into a practice.

Module 3: Contributions to the wider research and innovation community

Collective response under time pressure. My clearest contribution to the research community came during the COVID-19 pandemic, when I worked within collaborations assembled faster than research collaborations normally are. I contributed analysis to a seroprevalence study in an urban district of Cameroon published in *Nature Communications* (2021), and to an international effort to support the sub-Saharan African COVID-19 response by pooling analytical capacity across institutions, published in *BMC Public Health* (2022). The second of these is, in a sense, a paper about the research community itself: what it takes to make distributed analytical capacity usable in an

emergency. I also built and maintained an epidemiological dashboard for the WHO African Region and contributed to regional incidence analysis published in *Epidemiology & Infection*.

Closer to home, I contributed to age-structured scenario modelling for the Swiss SARS-CoV-2 epidemic (*F1000Research*, 2020) at a point when cantonal and federal decision-makers had little else to work from. That experience - producing projections known to be fragile, for people who had to act on them anyway - is the direct origin of the research question I now propose to answer.

International consortia. I currently contribute to two: **GESICA** (Interreg VI France–Switzerland), where I work with HUG emergency medicine, the Université Marie et Louis Pasteur, HES-SO and industrial partners on decision support for health crises across the Franco-Swiss border region; and a **Horizon Europe consortium** on artificial intelligence for epidemic intelligence, where my contribution concerns large language models for knowledge extraction, alongside partners including CIRAD, INRAE and the Istituto Superiore di Sanità. Both positions are as a contributor. I have found the experience valuable and also clarifying: I know now what I want to lead, and it is not the same thing as what these consortia build.

Open tools and reproducibility. I release what I build. LiteRev and the LiteRev-Evidence platform are available for use by other researchers, and the methodological work I propose is planned around open software and an openly released benchmark dataset. My view is that in fields where methods are the contribution, an unusable method is not really a contribution - a position that comes from working in an industry where a model nobody could run was a model nobody used.

Peer review and scientific exchange. I review occasionally for Springer and PLOS journals, less often than I am asked, and I give priority to reviews in my own methods area. I have presented this work at scientific meetings including BioTechX Europe, the International Workshop on HIV Observational Databases, the UNIGE Data Science Day and a CODATA/DDI Alliance workshop.

Bridging communities. My background places me between quantitative finance, econometrics and public health, and a recurring contribution has been importing methods across that boundary - extreme value estimation, regime models, decision-analytic evaluation - into a field where the underlying problems are structurally similar and the tools are largely unknown.

Module 4: Impact beyond the scientific community, for broader society

Evidence for cantonal public health, with funding I helped secure. Since 2025 I have led the data work on a study linking confirmed legionellosis cases in Geneva to individual domestic hot water installations, with technical, meteorological and territorial covariates. The question is practical and current: Switzerland is lowering hot water temperatures to meet energy targets, *Legionella* proliferation is temperature-dependent, and the trade-off has not been quantified with linked real-world data. Building it required geocoding cases, matching them to installation records, and constructing a linkage that to my knowledge exists nowhere else. Ethics approval was granted (BASEC 2026-00324), and the results and prioritisation tool go to the cantonal doctor to support surveillance and prevention planning.

What I would highlight is not the analysis but the assembly. The study is funded by partners with different reasons to care and different constraints - **Services Industriels de Genève and the Office cantonal de l'énergie** (CHF 30,000 each, obtained), with a further CHF 30,000 requested from the **Service du Médecin Cantonal** - a utility, an energy regulator and a public health authority. I co-wrote the funding applications. Bringing the partners to a common protocol, and satisfying the data-protection requirements of all three, was most of the work.

Emergency preparedness with the people who respond. Through GESICA I work directly with HUG emergency medicine and the Geneva emergency response system on decision support for health crises. The orientation of that work, and of the project I am proposing, is towards the users rather than the literature: the escalation thresholds that matter are the ones at which an emergency service can actually do something different, and those can only be established by asking the people who would do it.

Pandemic response. During COVID-19 I built and maintained an epidemiological dashboard for the WHO African Region, used for regional situational awareness, and contributed to modelling that informed Swiss federal and cantonal planning. This was work whose value was measured in whether it was usable in the week it was needed, not in citations, and it durably changed what I consider a successful methodological contribution.

Teaching practitioners. I lecture in statistics and epidemiology on the MAS in Public Health, a programme largely attended by practising clinicians and public health professionals. The transfer here is direct: methods reaching people who apply them in cantonal and hospital settings.

Why this matters for the work I want to do next. A recurring theme across all of the above is that technically sound analysis frequently fails to change anything, because it was

never designed to. The project I am proposing takes that failure as its object: it evaluates crisis forecasts on whether they alter decisions and outcomes rather than on statistical accuracy, and it treats the elicitation of decision thresholds from emergency responders as a research contribution rather than an implementation detail.

Research output list: 10 items

`[VERIFY]` Assumed cap: **10 works in total** across all modules, distributed as you choose. Confirm against the SNSF CV guidelines. Verify every bibliographic detail against the publisher record before submission.

Contribution statements matter more than author position. Conventions differ across fields and panels cannot infer your role. Each entry below states it explicitly, in one line.

Module 1: Generation of new ideas, tools, methodologies or knowledge (4)

1. Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. An automated literature review tool (LiteRev) for streamlining and accelerating research using natural language processing and machine learning. *Journal of Medical Internet Research* 2023;25:e39736. > *Contribution:* First author. Conceived the tool, designed and implemented the NLP, > dimensionality-reduction and clustering pipeline, conducted the performance evaluation, wrote > the manuscript. Funded by UNIGE CHF 30,000, the Fondation Ernst et > Lucie Schmidheiny CHF 10,000, Venture Kick CHF 10,000, Mimosa CHF 20,000). Now used by other > groups, including in the systematic review at item 5.

2. Orel E. LiteRev-Evidence: a platform for structured quantitative evidence extraction and scenario modelling. Software and research resource, 2024–. literev-scenario.com > *Contribution:* Sole architect and developer. The full review pipeline in working form: > live federated search of the open literature from natural-language or Boolean queries, > screening, living reviews, provenance-tracked and quality-weighted extraction pooled into > parameter distributions, and the modelling components that consume them. > `[[Consider depositing a versioned release with a DOI (Zenodo) before submission - a citable > identifier makes a software output count as an output rather than as a claim.]]`

3. Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa. *PLoS ONE* 2022;17(3):e0264429. > *Contribution:* First author. Designed the analysis, implemented the gradient-boosted models, > conducted the cross-country transportability assessment, wrote the manuscript.

4. Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. Difference in clinical severity between Delta variant and Omicron sub-lineages in a Swiss tertiary center. *CMI*

Communications 2024. > *Contribution:* First author. Designed and conducted the analysis of hospital clinical data, > wrote the manuscript.

Module 2: Development of other individuals and research teams (2)

5. Ng'ambi W, Estill J, Keiser O, Merzouki FA, **Orel E**. Machine learning-based classification of self-reported cardiovascular disease history in Africa using harmonised multi-country WHO STEPS surveys: 2014–2019. **Accepted for publication in *BMC Global and Public Health*, 2026.** Preprint: medRxiv, doi:10.64898/2026.02.23.26346870 > *Contribution:* **Last (senior) author.** Supervised the statistical analysis and the code. > Senior-authoring the work of the first author marks the transition from contributing to > collaborative analyses to directing them.

6. Edjinedja KL, Larribau R, **Orel E**, Cossus J, Elfahim O, Yendouname K, Vaussenat F, Teodoro D, Robert-Nicoud S, Barakat O, Desmettre T. Artificial intelligence in emergency medical services for disasters and health emergencies: a systematic review. Submitted to *Machine Learning and Knowledge Extraction*, 2026. > *Contribution:* Third author. Provided the methodological infrastructure (LiteRev, used in the > published methods to structure the review corpus) and guided its application through > screening and clustering, working with the doctoral researcher leading the review.

Module 3: Contributions to the wider research and innovation community (2)

7. Nwosu KD, Fokam J, Wanda F, Mama L, **Orel E**, Ray N, Meke J, Tassegning A, Takou D, Mimbe E, Stoll B, Guillebert J, Comte E, Keiser O, Ciaffi L. SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon. *Nature Communications* 2021;12:5851. doi:10.1038/s41467-021-25946-0 > *Contribution:* Senior statistician mentoring the analysis team, including the sample size > calculation and the analysis plan.

8. Estill J, Venkova-Marchevska P, Roelens M, **Orel E**, Temerev A, Flahault A, Keiser O. Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model. *F1000Research* 2020;9:646. doi:10.12688/f1000research.24497.2 > *Contribution:* Collaborated on the analysis and the coding of the age-structured scenario > model produced for Swiss pandemic planning under time pressure.

Module 4: Impact beyond the scientific community (2)

9. Orel E. COVID-19 epidemiological dashboard for the WHO African Region. Research resource, 2020–2021. > *Contribution:* Designed and built the dashboard (R Markdown) used for regional situational > awareness across the WHO African Region during the pandemic. > The dashboard is no longer online, no public archive exists.

10. Orel E (data lead), Keiser O (project leader). Legionellosis in Geneva: modelling of risk factors, geospatial and temporal analysis, and development of a predictive tool. Study protocol and linked dataset, BASEC 2026-00324, Commission cantonale d'éthique de la recherche, Geneva, 2026. > *Contribution:* Designed and built the linkage between the cantonal legionellosis case registry > and individual hot water installation records - a dataset with no equivalent elsewhere - and > co-wrote the funding applications (CHF 30,000 each from Services Industriels de Genève and the > Office cantonal de l'énergie, CHF 30,000 requested from the Service du Médecin Cantonal).

Notes on what was left out, and why

Two manuscripts in the pipeline are deliberately not listed while unpublished: the **first-author** *BMJ Open* submission (Place, pathway, and psychosocial distress among Nigerian HIV key populations starting ART, under review, July 2026) and - unless accepted or preprinted by submission - item 5 above has its documented swap-back. If the *BMJ Open* paper is **accepted before the SNSF deadline**, reconsider the allocation: a fourth published first-author paper may then earn a slot over item 7 or 8.

Roughly fifteen further peer-reviewed papers are not listed, including work on HIV in Malawi (*BMJ Paediatrics Open*; *Tropical Medicine & International Health*), sociobehavioural clustering (*PeerJ*; *Journal of the International AIDS Society*), treatment-interruption prediction (*JAIDS*), tuberculosis burden (*Lancet Regional Health – Western Pacific*) and semi-automated review (*JMIR*).

That omission is a deliberate claim. The ten selected outputs trace one line - methods and tools for prediction and evidence synthesis, applied to public health problems, increasingly led by me - rather than sampling the record evenly. A narrative CV rewards judgement about your own work, and a list that showed everything would show no judgement at all.

Two decisions worth revisiting before you submit:

- **Item 2 (the platform) occupies a slot that could hold a peer-reviewed paper.** I think it earns its place: it is the instrument the whole proposal rests on, it is unambiguously yours, and it is the strongest available evidence that the project is feasible. But it is the entry a conservative reviewer is most likely to discount, and a Zenodo DOI would substantially reduce that risk.
- **The Ng'ambi CVD paper (item 5) is accepted for publication** (status confirmed 19 Aug 2026). The earlier swap-back contingency is closed - it holds its slot as the strongest Module 2 entry. Update venue, year and DOI when they are assigned.